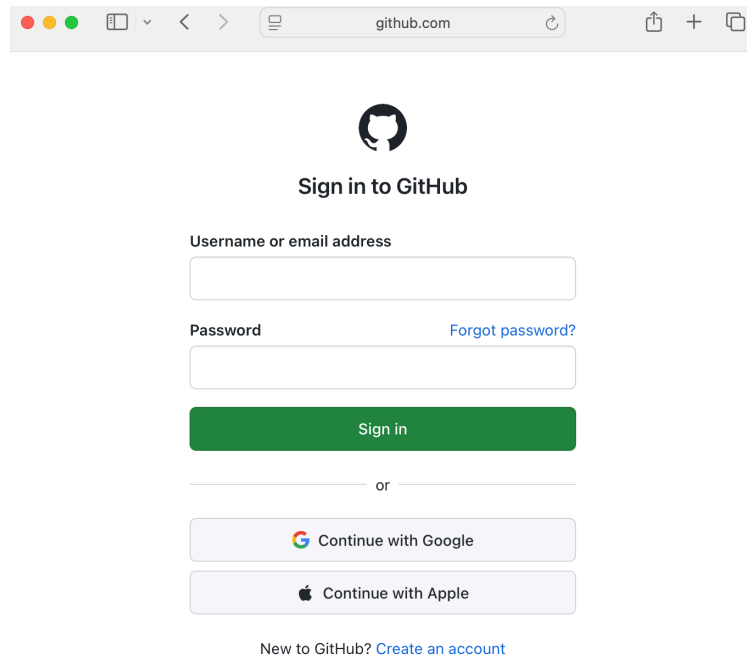


Installing Software for Robotics

1. Github

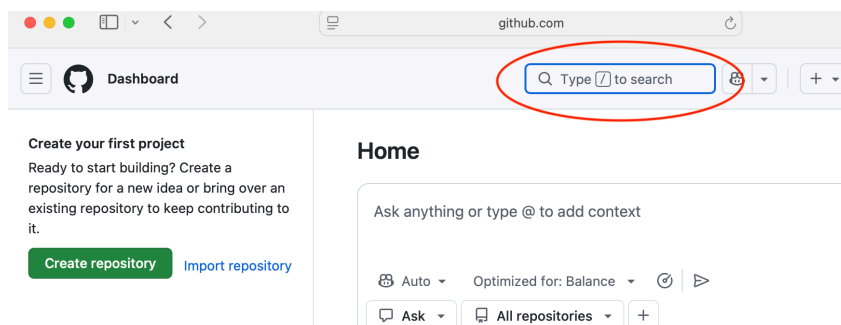
Github is an online platform where we can keep common code to share within teams. That way we can all work on the same code.

- In a web browser (Safari, Edge, etc.), go to <https://github.com> and log-in or create an account.



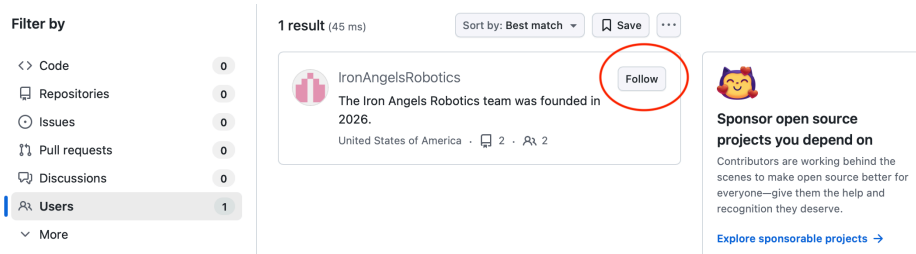
A screenshot of the GitHub login page in a web browser. The browser's address bar shows 'github.com'. The page features the GitHub logo at the top, followed by the text 'Sign in to GitHub'. Below this are two input fields: 'Username or email address' and 'Password'. A green 'Sign in' button is positioned below the password field. To the right of the password field is a link that says 'Forgot password?'. Below the 'Sign in' button is a horizontal line with the word 'or' in the center. Underneath this line are two buttons: 'Continue with Google' and 'Continue with Apple'. At the bottom of the page, there is a link that says 'New to GitHub? Create an account'.

- In the Homepage, search for IronAngelsRobotics by typing it in the search bar

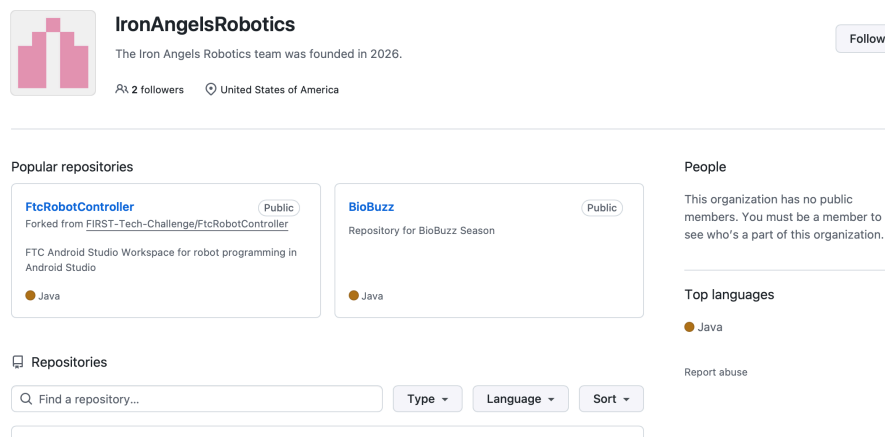


A screenshot of the GitHub homepage in a web browser. The browser's address bar shows 'github.com'. The page has a header with a 'Dashboard' link and a search bar. The search bar is highlighted with a red circle and contains the text 'Type / to search'. Below the header, the page is divided into two main sections. The left section is titled 'Create your first project' and contains a green 'Create repository' button and a blue 'Import repository' link. The right section is titled 'Home' and contains a search bar with the text 'Ask anything or type @ to add context'. Below this search bar are several filters: 'Auto', 'Optimized for: Balance', 'Ask', and 'All repositories'.

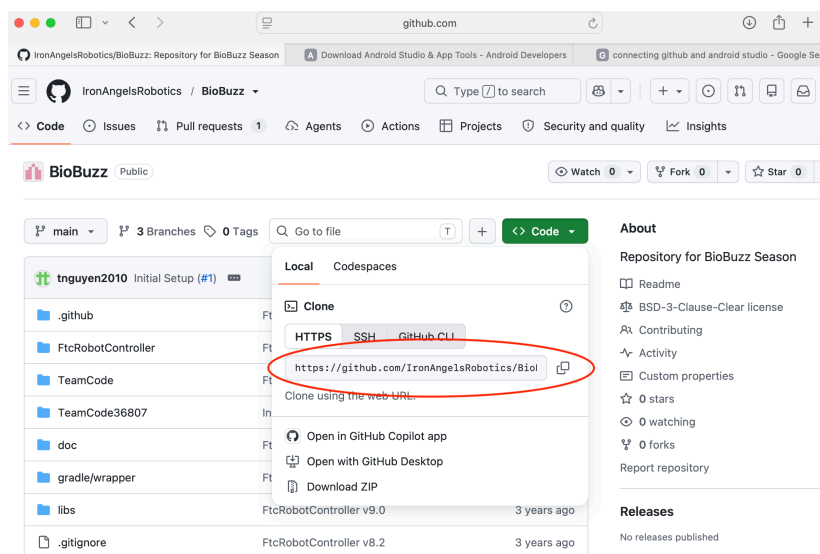
- You will find the group after clicking on the Users link in the list on the left side of the results page. Click the Follow button on the IronAngelsRobotics result.



- Then, click on the IronAngelsRobotics link (The name, next to the icon) to go to our team's GitHub page. You will find two repositories (projects, or code groups) there. Click on BioBuzz.



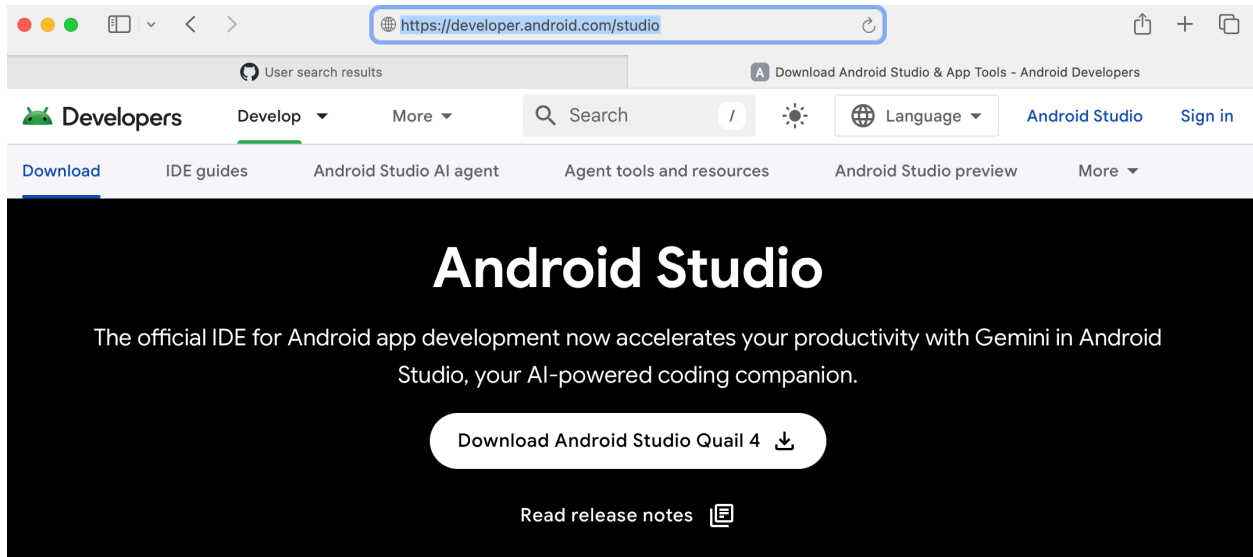
- On the BioBuzz repository page, click on the green Code button. This takes you to the repository's web address. You will need it later in step 4.



2. Android Studio

Android studio is the tool that we will use to write code and load it to the robot.

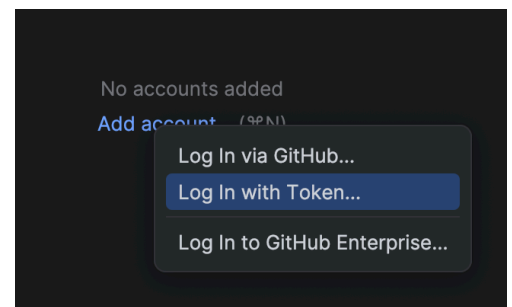
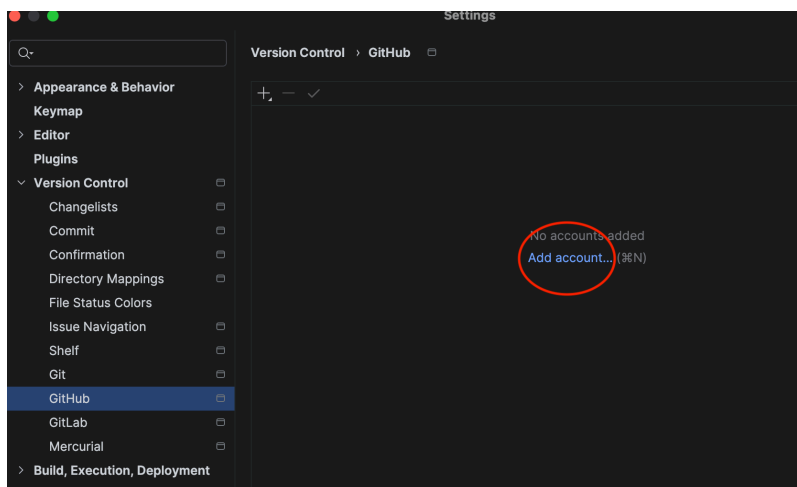
- Go to <https://developer.android.com/studio> and click the Download Android Studio button and install it after it downloads to your computer (Use standard installation settings when asked)



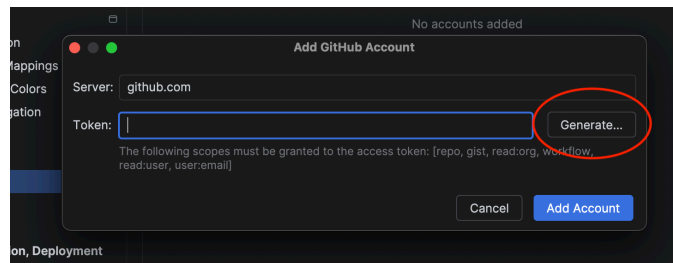
3. Connecting Github to Android Studio

This is needed to share code: access what everyone is working on, and make contributions.

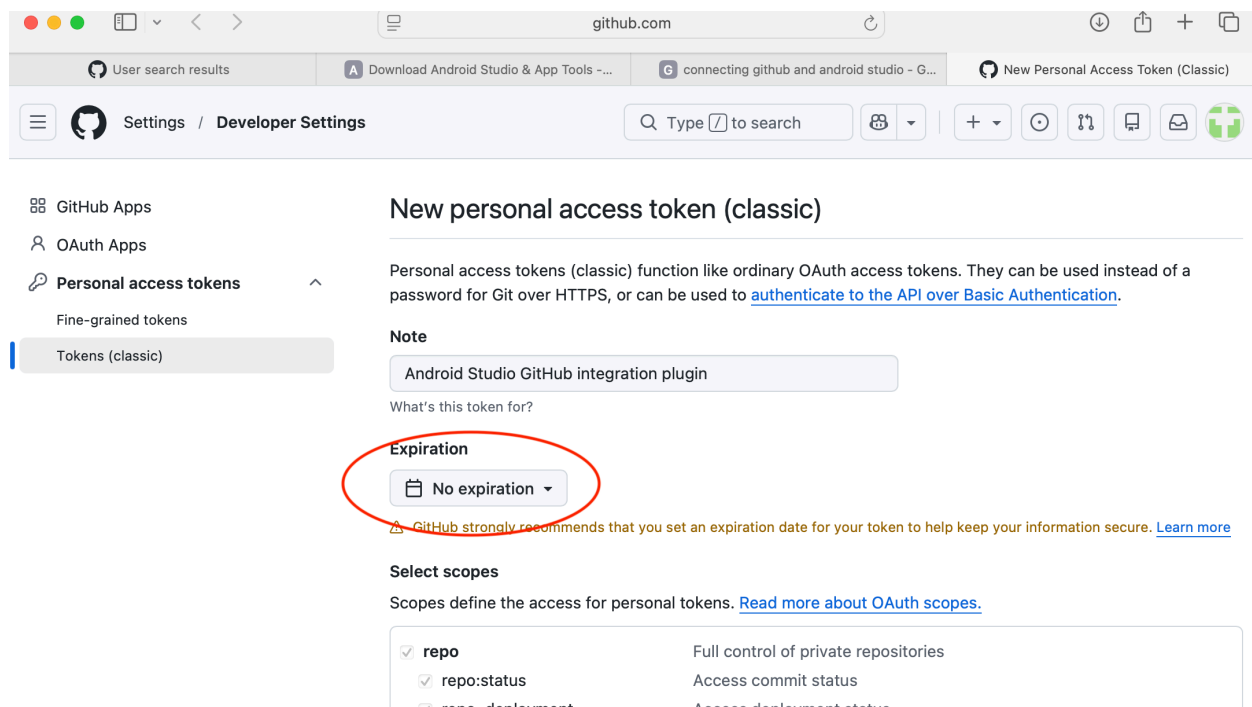
- Go to File > Settings (Windows) or Android Studio > Settings (macOS)
- Navigate to Version Control > GitHub
- Click the + (Add) icon and select Log In with Token... [1]



- Click Generate... to open GitHub in your browser and log in using your GitHub account from step 1. Note: if you are still logged in from step 1, you will not need to log in.



- In the window that appears after logging in to GitHub, select “no expiration date” and click the green “Generate Token” button at the bottom.



☐ read:project	read access of projects
☐ admin:gpg_key	Full control of public user GPG keys
☐ write:gpg_key	Write public user GPG keys
☐ read:gpg_key	Read public user GPG keys
☐ admin:ssh_signing_key	Full control of public user SSH signing keys
☐ write:ssh_signing_key	Write public user SSH signing keys
☐ read:ssh_signing_key	Read public user SSH signing keys

Generate token Cancel

- Then, copy the generated token (a long string of numbers and letters) by selecting it, or clicking the copy button next to it. Then, paste other into the Token field in Android Studio and click the Add Account button.

Personal access tokens (classic)

Generate new token ▼

Tokens you have generated that can be used to access the [GitHub API](#).

 Make sure to copy your personal access token now. You won't be able to see it again!

✓ ghp_6sA [REDACTED] a3N5h  Delete

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Add GitHub Account

Server:

Token: Generate...

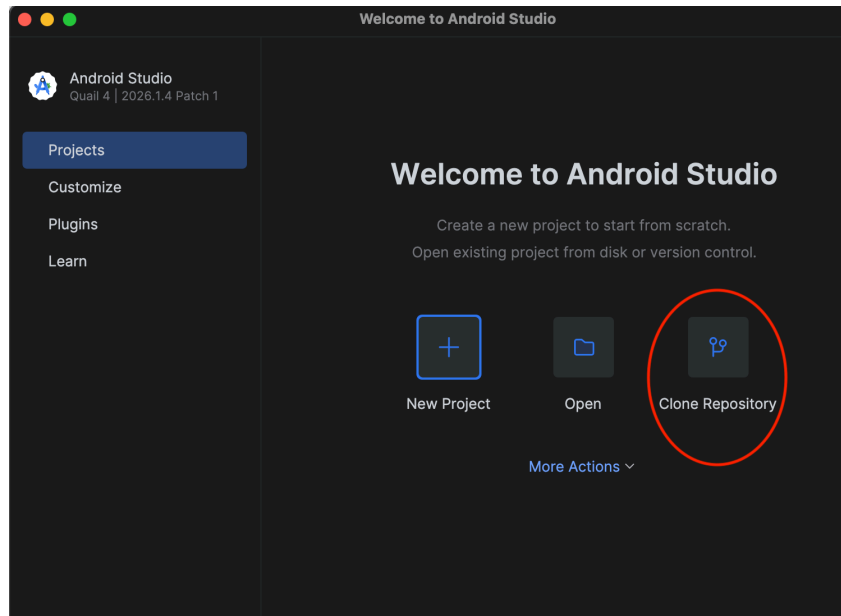
The following scopes must be granted to the access token: [repo, gist, read:org, workflow, read:user, user:email]

Cancel Add Account

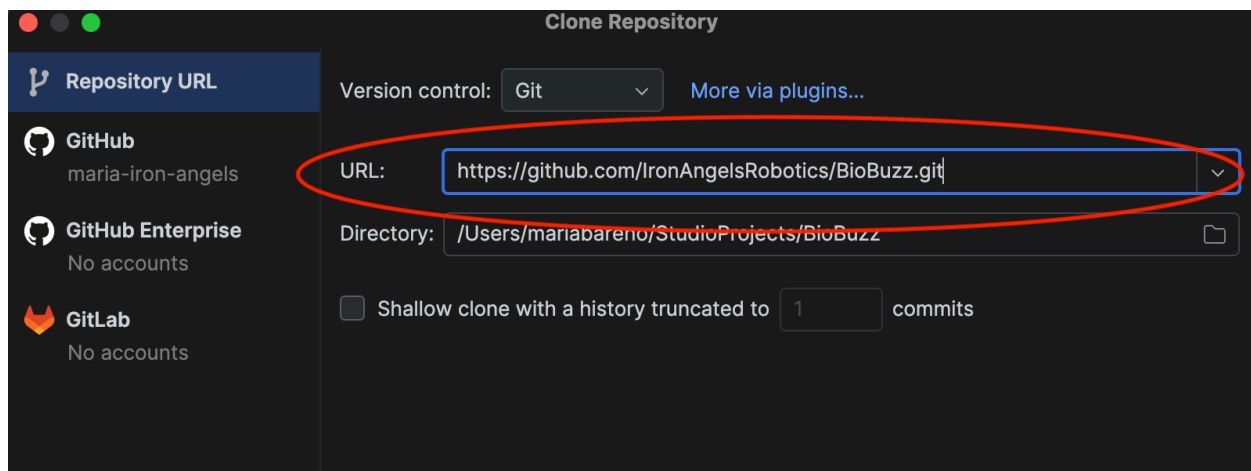
4. Cloning the Repository

Cloning a repository is getting a copy of the shared code we can work with. But this copy will stay connected to the shared version, so we can get changes and share ours.

- In Android Studio, select Clone Repository (or File > New > Project from Version Control)



- Paste the BioBuzz web address from step 1 into the URL field and click the blue Clone button at the bottom right side of the window. When asked, select “trust this folder” or “trust this repository”.



- Now you have a copy of the team's code to play with. And you will be able to get more code that we add, and add your own contributions. If you want to check the code we used in the meeting to run the motor and spin the wheel, it is under the TeamCode36807 folder (we are Team 36807)

